changing the hardware address (ethernet MAC address, for example) that an IP address resolves to.

To enable TLS, you'll need to generate certificates for each service and distribute the certificates and private keys to each pod that implements that service. The processes involved in creating and distributing certificates is known as *public key infrastructure* (PKI).

DEFINITION A ***public key infrastructure*** is a set of procedures and processes for creating, distributing, managing, and revoking certificates used to authenticate TLS connections.

Running a PKI is complex and error-prone because there are a lot of tasks to consider:

- Private keys and certificates have to be distributed to every service in the network and kept secure.
- Certificates need to be issued by a private certificate authority (CA), which itself needs to be secured. In some cases, you may want to have a hierarchy of CAs with a *root CA* and one or more *intermediate CAs* for additional security. Services which are available to the public must obtain a certificate from a public CA.
- Servers must be configured to present a correct certificate chain and clients must be configured to trust your root CA.
- Certificates must be revoked when a service is decommissioned or if you suspect a private key has been compromised. Certificate revocation is done by publishing and distributing *certificate revocation lists*